A Training-Free Pipeline for Story-Aware Audio Description: Vocal-Isolated ASR Context and Character-Grounded Prompting

SLoMO-AD Competition 2026 — Technical Report

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Code: https://github.com/ZFTurbo/SLoMO-AD-Competition-2026

Abstract

This report describes a submission to the Audio Description (AD) Generation track of the SLoMO Competition at ECCV 2026. The solution is entirely *training-free*: no component is fine-tuned on the competition data. The pipeline enriches each target time interval with two auxiliary signals that a video-only model does not have access to — (i) a dialogue transcript obtained by first separating the vocal stem from the mixed soundtrack and then running ASR on the isolated voice, and (ii) a set of character reference portraits with their in-film names — and passes both, together with the video segment itself, to a large multimodal model under a hard length budget derived from the duration of the dialogue gap. The same pipeline serves both tracks, differing only in the backbone: Qwen3.8-27B-FP8 [13] in reasoning mode for the Main Track and Qwen3-VL-8B-Instruct [14] for the Special (lightweight) Track. On the public test leaderboard the two configurations reach an AD score of 50.07 and 46.83 respectively, with the top reported entry at 54.49. The whole system runs on a single 96 GB GPU.

1 Introduction

Audio Description is a spoken narration inserted into the pauses between dialogue so that blind and low vision viewers can follow a film. The competition provides a movie and a set of target intervals (the dialogue gaps) and asks for one description per interval. Prior work on this task has been predominantly training-based, pairing a video encoder with a language model fine-tuned on aligned AD corpora [3, 2, 1]. Relative to generic video captioning, the task adds three requirements: descriptions must be *concise* enough to fit the gap, *selective* in reporting only story-critical visual content, and *story-aware*, tracking named characters rather than describing frames in isolation. Submissions are ranked by a single AD score combining ADQA (a question-answering measure of story content) and CIDEr [6]. The two evaluation sets, CMD-AD [1, 4] and SF20K [5], differ sharply in character: the former consists of clips from commercial films with professionally narrated reference ADs, the latter of amateur short films chosen specifically to minimise overlap with web-scale pretraining corpora.

The design follows from a simple observation about where a general-purpose multimodal model fails on this task. Given only a short clip, such a model produces fluent but generically phrased captions: it says "a man" where the reference AD says "Cobb", and it re-narrates events that the viewer can already hear. Both failure modes are addressable without training. Character identity can be supplied through the provided charbank as visual references paired with names, and redundancy with the soundtrack can be suppressed by showing the model what is being said, while explicitly instructing it to describe only what is visible. The remaining requirement, conciseness, is a formatting constraint that can be imposed directly through the prompt.

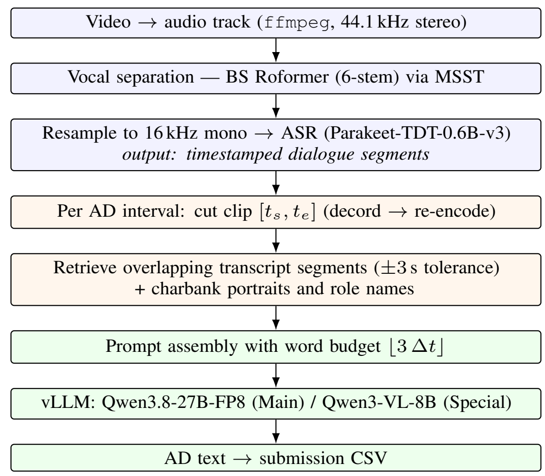

Figure 1: Overview of the pipeline. The blue block is run once per movie offline; the orange and green blocks are run once per AD interval.

Accordingly, the whole system is training-free and consists of an offline audio preprocessing stage and an online inference stage. This keeps the method portable across the two tracks: only the backbone changes between the Main and Special submissions, while the preprocessing artefacts are shared.

2 Method

Figure 1 summarises the pipeline. Each stage is described below.

2.1 Audio preprocessing

Track extraction. For every video in both datasets the audio track is demuxed with FFmpeg [18] to 16-bit PCM at 44.1 kHz, stereo, without re-encoding the video.

Vocal separation. Movie soundtracks mix speech with music, effects and ambience, and transcribing this mixture directly yields noisy hypotheses and unreliable timestamps — precisely the two properties the pipeline depends on later. The vocal stem is therefore separated before recognition, using the BS-RoFormer [7] 6-stem model through the Music-Source-Separation-Training (MSST) framework [8, 19] and keeping only the `vocals` output. This step is the single most useful preprocessing decision in the pipeline: it removes the score and the effects bed, which otherwise dominate loud action passages where dialogue is sparse and AD intervals are frequent.

Speech recognition. The isolated vocals are downmixed to 16 kHz mono and transcribed with NVIDIA `parakeet-tdt-0.6b-v3` [9], a FastConformer encoder [10] with a token-and-duration transducer decoder [11] trained on the Granary corpus [12]. Since full films exceed the model's practical context, audio is split into 300 s chunks, transcribed independently, and the returned word- and segment-level timestamps are shifted by the chunk offset before being concatenated. The stage emits, per movie, a CSV of `(start, end, segment)` triples that is cached on disk and reused by both tracks.

2.2 Per-interval clip construction

For an AD interval $[t_s, t_e]$ with duration $\Delta t = t_e - t_s$ (clamping $t_s \geq 0$), the source video is opened with `decord` [17], the frame range $[\lfloor t_s f \rfloor, \lfloor t_e f \rfloor]$ is sliced at the native frame rate f, and the slice is re-encoded to a temporary MP4 at the original frame rate. Temporary clips are deleted immediately after the corresponding request completes, so peak disk usage stays constant.

Rather than fixing a sampling rate, the pipeline fixes a frame *budget*: the serving-side sampling rate is set to

$$f_{\text{target}} \;=\; \min\!\left(\frac{N}{\Delta t},\; f\right), \qquad N = 50,$$

so that every clip contributes approximately N frames to the model regardless of its length. Short gaps are thus sampled densely and long gaps sparsely, which keeps the visual token count — and therefore latency — roughly constant across a highly variable interval-length distribution. Frames are additionally bounded to at most 360×420 pixels of processed resolution.

2.3 Character grounding

For each movie the provided charbank is read and the portraits of its characters are collected, keeping at most 26 entries and silently skipping characters whose image file is absent. The two datasets are indexed differently — CMD-AD resolves characters through the `imdbid` of the clip and stores portraits under actor identifiers, whereas SF20K stores an explicit image path per character — but both are normalised to the same list of `(image, name)` pairs. The images are passed to the model as an ordered image sequence, and the prompt enumerates them explicitly ("Image i is the character '*name*' "), so that identity is established by position rather than by any assumption about the model's prior knowledge of the cast. This is what makes named references possible on SF20K, whose amateur short films are unlikely to appear in web-scale pretraining data.

2.4 Dialogue context retrieval

A transcript segment $[s, e]$ is attached to the interval $[t_s, t_e]$ if it overlaps the interval, or if it begins within 3 s after the interval ends, or ends within 3 s before it begins. The asymmetric tolerance is deliberate: an AD is spoken in a gap between utterances, so the lines immediately *surrounding* the gap usually carry more information about what is happening on screen than anything inside it. The retrieved lines are rendered as a bulleted block. When the retrieval returns nothing — silent passages, or films where separation and ASR produced no output — a shorter prompt variant without the dialogue section is used instead, which avoids presenting the model with an empty context block that it may try to explain.

2.5 Length control

Descriptions must fit the gap. Duration is converted to a word budget at a speaking rate of three words per second,

$$W \;=\; \max(\lfloor 3\,\Delta t \rfloor,\; 2)\,,$$

and W is stated in the prompt as a hard requirement. This also serves the qualifying AD Duration Check. In the Special Track the budget is additionally enforced structurally by capping generation at 128 tokens.

2.6 Prompt

The full instruction, with the dialogue context present, is:

```
You are given {n_actors} reference images
of characters, one main video, and a
transcript of the dialogue occurring in
and around this video clip.
--- CHARACTER REFERENCES ---
{actor_references}
--- DIALOGUE CONTEXT ---
{dialogues_text}
--- INSTRUCTIONS ---
Please provide a detailed visual
description of the main actions, subjects,
and scene changes IN THE VIDEO. Use the
dialogue context only to better understand
the situation, but describe only what
is visibly happening.  When describing
people, if you visually recognize any
of the referenced characters, use their
exact names.  Crucially, if a referenced
character is NOT visible in the video,
do NOT mention them at all (even if they
are speaking in the dialogue).  CRITICAL
REQUIREMENT: Your entire response MUST be
strictly under {words_count} words!
```

Two clauses were added in response to observed failures. Without the instruction to use the transcript only for disambiguation, models paraphrase the dialogue instead of describing the image, which is doubly penalised: it duplicates information the viewer can already hear and it consumes the word budget. Without the explicit prohibition on mentioning absent characters, models introduce off-screen speakers into the description, or — worse — state that a character is *not* present, which is never a valid AD.

2.7 Models and serving

Both backbones are served with vLLM [16] behind an OpenAI-compatible endpoint, with local media access enabled so that clips and portraits are passed by file path rather than base64, and with the multimodal processor configured for per-request `fps`.

Main Track. Qwen3.8-27B-FP8 [13] in reasoning mode, with `temperature` $= 1.0$, `top_p` $= 0.95$, no presence penalty, a generation limit of 32,768 tokens and the highest reasoning effort setting; the model inherits the hybrid-attention and thinking-mode design of the Qwen3 line [15]. Thinking is enabled and preserved in the response, and the visible answer is recovered by splitting on the closing think delimiter. The long generation limit is required by the reasoning trace, not by the answer, which remains bounded by W.

Special Track. Qwen3-VL-8B-Instruct [14] (8B dense, within the $<$10B activated-parameter limit), sampling with default settings and `max_tokens` $= 128$.

3 Implementation Details

All runs were performed on a single NVIDIA A6000 Blackwell 96 GB GPU. The vLLM server is launched with a 131,072-token context, FP8 weights for the 27B model, and a GPU memory utilisation of 0.95; Results are cached per movie: a completed movie writes a CSV that is reloaded verbatim on restart, and every individual request is additionally dumped with its full prompt for inspection. This makes long runs interruptible and made prompt-level debugging tractable, since a regression can be traced to the exact multimodal payload that produced it.

The two datasets are handled by separate scripts with identical logic and differing only in path and metadata conventions, which was preferred over a single parameterised script because the charbank schemas do not align.

4 Results

Table 1 reports the public test leaderboard AD score.

Model	Track	AD score
Qwen3.8-27B-FP8 (reasoning)	Main	50.07
Qwen3-VL-32B-Instruct-FP8	Main	—
Qwen3-VL-8B-Instruct	Special	46.83
Top reported entry	Main	54.49

Table 1: Public test leaderboard results. "—" denotes a configuration evaluated internally but not submitted separately.

Two findings are worth reporting. First, Qwen3-VL-32B-Instruct-FP8 [14] scored slightly below the 27B reasoning model but ran substantially faster, which makes it the better choice under a wall-clock constraint; the reasoning trace of the Main Track model is by far the dominant cost of the pipeline, and it buys a modest margin. Second, the 8B model in the Special Track is only about three points behind the Main Track submission despite roughly a third of the parameters and no reasoning. This is evidence that, for this task, most of the achievable gain at the current operating point comes from *context* — character identity, dialogue, and an explicit length budget — rather than from backbone capacity. It also suggests the gap to the top entry is unlikely to close by scaling the backbone alone.

5 Limitations and Future Work

The system has no memory across intervals. Each AD is generated independently, so the pipeline satisfies character grounding but only approximates story-awareness: it cannot avoid repeating an establishing detail already narrated a minute earlier, nor refer back to an event it described before. Carrying a running summary of previously emitted descriptions into the prompt is the most promising next step, and is compatible with the training-free design.

Character grounding degrades when a movie has a large cast, since the portrait list is truncated and all remaining references compete for attention within one context; retrieving only the characters plausibly present in the clip would be a better use of the budget. The word budget is a proxy for duration and does not account for how a specific sentence would actually be spoken. Finally, the pipeline inherits the errors of its front end: separation artefacts and ASR errors propagate silently into the dialogue context, and the retrieval tolerance is a fixed constant that was not tuned.

6 Reproducibility

The repository contains the complete pipeline. After placing the private test data under `./input/`, preprocessing is run once with `python preproc_data.py ./input/`; the BS Roformer checkpoint is fetched automatically on first use. Inference then requires a running vLLM server for the selected backbone, after which the two per-dataset scripts are executed for the chosen track. The exact server invocations and argument values are given in the repository README. No component is fine-tuned, and no competition training data is used.

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